

File Permissions/Mode

- File permission types: r (read), w (write), x (execute)
- Permission levels: u (user), g (group), o (other)
- File mode: u-rwx g-rwx o-rwx
- Example: hello.txt -- user can read/write and execute, group can read and execute, other can only read.
 - hello.txt mode: u=rwx g=r-x o=r--
 - rwx r-x r--
 - 111 101 100 = 754
 - chmod 754 hello.txt

Directory permissions

- r (read): Can read directory entries.
 - ls command can list the directory.
 - readdir() can get directory entry.
- w (write): Can add new directory entry, modify existing directory entry, or delete directory entry.
 - new directory entry: when create new sub-directory (e.g. mkdir) or file (e.g. touch, ...).
 - modify existing directory entry: rename file or sub-directory.
 - delete directory entry: delete file or sub-directory (e.g. rm, rmdir, ...)
- x (execute): Can browse the directory
 - cd command (chdir() syscall) will work

chmod

- chmod permissions filename
- Permissions:
 - +x/r/w - add permission to all
 - -x/r/w - remove permission of all
 - u+x/r/w - add permission to user
 - u-x/r/w - remove permission of user
 - g+x/r/w - add permission to group
 - g-x/r/w - remove permission of group
 - o+x/r/w - add permission to others
 - o-x/r/w - remove permission of others
 - 0774 - apply rwx to user, rwx to group and r to others

chown

- sudo chown username filename
- sudo chown :groupname filename
- sudo chown username:groupname filename

chgrp

- sudo chgrp groupname filename

adduser

- `sudo adduser username`

passwd

- `sudo passwd username`

deluser

- `sudo deluser username`
 - home directory is not removed
- `sudo deluser -r username`
- `sudo deluser --remove-home username`

Links

- Links are shortcuts to access deeply located files quickly.
- There are two types of links in Linux/UNIX.
 - 1. Symbolic Link
 - 2. Hard Link

Symbolic Link

- `terminal> ln -s /path/of/target/file linkpath`
- Internally use `symlink()` syscall.

symlink() syscall

- A new link file is created (new inode and new data block is allocated), which contains info about the
 - target file (absolute or relative path).
- Link count is not incremented.
- If target file is deleted, the link becomes useless.
- Can create symlinks for directories also.

Hard Link

- `terminal> ln targetfilepath linkfilepath`
- Internally use `link()` syscall.

link() syscall

- A new directory entry is created, which has a new name and same inode number. No new file (inode and data blocks) is created.
- Link count in the inode of the file is incremented.
- If directory entry of target file is deleted (rm command), file can be still accessed by link directory entry.
- Cannot create hard link for directories, because it may lead to infinite recursion (while traversing directories recursively e.g. `ls -R`)

VI Editor

- `sudo apt-get install vim`

- VI editor works in two modes
 - command mode
 - insert mode
- press i - to go into insert mode
- press Esc - to go into command mode
- VI editor commands:
 - :w - write/save into file
 - :q - quit vi editor
 - :wq - save and quit
 - :y - to copy current line
 - :ny - to copy nth line
 - yy - to copy current line
 - nyy - copy n lines from current line
 - :m,ny - copy from mth line to nth line
 - :d - to cut current line
 - :nd - to cut nth line
 - dd - to cut current line
 - ndd - cut n lines from current line
 - :m,nd - cut from mth line to nth line
 - press p - to paste copied line on next line of current line
 - yw - copy from current position upto next word
 - yiw - copy current word
 - y\$ - copy from cursor position upto end of line
 - y^ - copy from cursor position upto start of line
 - vim -o - to open multiple files horizontally
 - vim -O - to open multiple files vertically
 - ctrl + ww - to go into next file
 - /pattern - to search the pattern
 - n - to go on next occurrence

- 😞/pattern1/pattern2/ - find and replace only first occurrence of current line
- 😞/pattern1/pattern2/g - find and replace all occurrences of current line
- :%s/pattern1/pattern2/g - find and replace all occurrences of file
- To do setting in vi editor
 - create file into home directory as below:

```
vim ~/.vimrc
```

- add below content into it

```
set number
set autoindent
set tabstop=4
set nowrap
```

Shell Scripting/Programming

- Shell Script - set of commands with some programming constructs, written for shell
- commands - shell commands
- Programming construct - control flow structures, functions, arrays
 - Decision control structure (if else)
 - Selection control structure (case)
 - Loop control structure (while, until, for)
- Shell scripts are interpreted by shell

Applications

- Shell scripts are used by administration people for installations, setups, automations
- Administrative tasks e.g. User creations, Troubleshooting, Router/Firewall config on commandline, etc.
- Automation of tasks e.g. Cron jobs, Scripts for building softwares, etc.
- Application installers

Limitations

- Slow execution -- Interpreted. Cannot be used for performance centric applications.

Procedure to write and execute scripts

```
* Step 1 - create file with extension .sh
* $ vim script.sh
* Step 2 - execute the script
* Option 1
```

```
* $ bash script.sh
```

```
* Option 2
```

```
* $ chmod +x script.sh
```

```
* $ ./script.sh
```

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